Abhimanyu Kanase

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Data & AI-Focused Working Student Candidate — Data Modeling, Generative AI, Agile Collaboration
Experienced in building cloud-based data models and dashboards within Agile/Scrum teams, using Generative AI tools such as ChatGPT and Claude alongside custom LLM systems to translate technical work into business-ready insights. Foundational knowledge of requirements engineering and cross-functional collaboration.

WORK EXPERIENCE

ML and Data Engineering Intern 10/2023 – 03/2024
Innovaters8teen India

- Maintained **data integrity** across **10,000+ labeled records**, measured by consistent validation pass rates, by building automated **categorisation and validation pipelines in Python.**
- Reduced **manual process work by ~80%** across **4+ recurring tasks**, measured by hands-on time saved, by automating **extraction workflows with Python and PowerShell.**
- Enabled **continuous quality monitoring**, measured by consistent automated test pass rates, by maintaining a **version-controlled codebase with CI/CD integration.**

Software Developer Intern (Python) 02/2023 – 05/2023
Insculpt Technologies LLP India

- Maintained **trustworthy reporting data** across **50,000+ rows**, measured by **zero downstream data-quality escalations**, by cleaning and validating **multi-source datasets** with **Python (Pandas) and SQL.**
- Enabled **faster, more confident business decisions**, measured by adoption across **3+ Power BI and Excel dashboards**, by translating raw data into clear **stakeholder-ready visualisations.**
- Eliminated **manual reporting effort**, measured by **~100% reduction in export time** across **3+ recurring workflows**, by automating **data-extraction and report-generation scripts.**
- Maintained **structured, organized documentation of data workflows**, measured by consistent reproducibility across reporting outputs, via **Git/GitHub.**

EDUCATION

Master of Science in Data Science 04/2025
Hochschule Fulda - University of Applied Sciences | Expected graduation: Late 2027 Germany

Bachelor of Engineering in Electronics & Telecommunication 06/2020 – 06/2024
Pimpri Chinchwad College of Engineering and Research | Final Grade: 1.9/5.0 India

PUBLICATIONS

Cotton Detection Using YOLOv5 for Agricultural Analysis 2024
IEEE, 5th International Conference on Emerging Technology (INCET 2024)

Computer Vision-Based Weed Killing Rover 2024
Springer, ICICCD 2024, Lecture Notes in Networks and Systems

SKILLS

Technical Skills

Data Modeling & Analytics: SQL, Python (Pandas, NumPy), dimensional/star-schema data modeling, Power BI, Looker Studio
Generative AI & LLMs: LangGraph, RAG pipelines, BERT fine-tuning, ChatGPT, Claude (productivity & analysis use)
Cloud Platforms: Microsoft Azure (Data Factory, ADLS, Synapse), AWS (S3, Athena), Snowflake, Databricks
Collaboration & Process: Jira, Agile/Scrum, Figma (UI/UX basics), requirement engineering fundamentals

Languages

English - Fluent
German – B1 (Currently Learning)

Soft Skills

Analytical Thinking,
Statistical Problem-Solving,
Attention to Detail,
Cross-functional Collaboration,
Presenting Findings to Stakeholders,
Independent Working Style

Interests

Photography, Video Editing

PROJECT EXPERIENCE

Agentic AI Chatbot with LangGraph & RAG

- Designed a modular **Generative AI agent system** with **LangGraph and FastAPI** to support **multi-step reasoning** across **1,000+ knowledge entries**, with **persistent conversation state** maintained across sessions.
- Reduced **context loss by ~30%** by building a **Retrieval-Augmented Generation (RAG) pipeline** using **vector embeddings and PostgreSQL**, improving response coherence across retrieved context.
- Enforced **consistent data standards** at the system boundary through **Pydantic-validated RESTful endpoints**, achieving **zero malformed request errors** during testing.
- Used **ChatGPT and Claude** regularly for **research, debugging, and analytical workflows**, combining **AI-assisted problem-solving** with **custom-built LLM systems** to accelerate development and iteration.

Automated Music Trend Pipeline (Airflow + dbt + PostgreSQL)

- **Replaced manual** data **handling** entirely, measured by **~100% reduction** in manual export work, by **automating daily** ingestion of **10,000+ records** via **Apache Airflow.**
- **Reduced** downstream **query** complexity by **~60%**, measured by **simplified**, **consistent query logic**, by building **3 dbt transformation models** (genre_summary, artist_trends, decade_analysis).
- **Established** a **self-monitoring data** foundation, measured by consistent, validated pipeline outputs, by applying structured data modeling practices across ingestion and transformation stages.
- **Enabled** continuous **stakeholder access** to fresh data, **measured** by **automated** dashboard **refresh cycles**, by connecting **pipeline output** to **Looker Studio.**

Netflix Data Analysis (Polars + Python)

- Established a reliable analytical foundation by cleaning and validating the dataset, identifying and resolving missing values and data quality issues with Polars and Python.

- Applied structured data science analysis techniques to identify actionable content trends across genre distribution, content ratings, and release-year data.
- Translated raw data into an accessible analytical narrative through a set of visualisations that clearly surfaced key findings and insights.
- Maintained a rigorous, quality-driven analytical process by applying consistent methodology from data ingestion through final output, with a detail-oriented focus on reliable results.

Cotton Detection & Harvesting Analysis (Computer Vision)
- Prepared, validated and piloted a YOLOv5-based ML model on 10,000+ large-scale field images, applying data augmentation and statistical feature engineering techniques to improve prediction accuracy by 15–20% — contributing to automated decision-making in agricultural operations.
- Conducted exploratory data analysis and model evaluation over 50+ training epochs using precision, recall and F1-score as statistical metrics, achieving ~85% precision and 0.80+ mAP — presenting findings and progress throughout development.
- Constructed a scalable, reproducible end-to-end ML pipeline in Python from large-scale data ingestion through preprocessing, model training, validation and deployment — documenting all development decisions independently.

Music Recommendation Engine using BERT (NLP / Music AI)
- Automated **genre prediction** across **52 categories**, measured by **precision/recall/F1 benchmarking**, by fine-tuning a **BERT-large Generative AI model** on **10,000+ track descriptions**
- Delivered clear, **decision-ready findings**, measured by consistent **evaluation metrics** presented across development milestones, by building structured **analyses and progress presentations**
- Built a full **NLP pipeline**, measured by successful **end-to-end runs from raw text to trained model**, by implementing **cleaning, feature extraction, and training stages**

Nike Website UI Clone (Personal Practice Project, 2024)
- Recreated the **Nike website interface in Figma** as a personal **UI/UX practice project** by analyzing and replicating its **component structure, spacing, visual hierarchy, and page layout** for a **high-fidelity design**.

Choosing the Nominal Correction for Laser Trimming of Platinum Thin-Film Temperature Sensors
- Served as **project lead for a 5-member team**, measured by **on-time delivery across all 7 project phases**, by coordinating **tasks, dependencies, and Agile workflows in Jira.**
- Reduced **projected production error rates**, measured by validated performance across iterative test cycles, by defining a final **physics-based correction method** through **forward and reverse modelling** — directly supporting lower **rework costs and improved product quality.**
- Selected the **most reliable model-selection approach** across **9 configuration cells**, measured by persistence outperforming alternatives in **6 of 9 cells**, by applying **temporal validation.**
- Delivered a **reproducible, cost-relevant production pipeline**, measured by **98.3% coverage of production orders**, by streamlining **Agile collaboration** across data, modelling, and deployment workstreams.

Spotify Streaming Analytics Dashboard
- Built a **unified analytics layer for business decision-making**, measured by successful consolidation of **10,000+ records across 3 sources**, by integrating **AWS S3, Athena, and BigQuery.**
- Surfaced **actionable business insights**, measured by **5+ documented findings** on audience and genre trends, by conducting structured **EDA in Python.**
- Delivered **stakeholder-ready reporting**, measured by adoption of **4+ dashboard visualisations**, by developing and maintaining a **Looker Studio dashboard.**

Natural Language Data Query API (FastAPI Microservice + LangGraph + Docker)
- Built a FastAPI microservice with Pydantic-validated endpoints processing 50+ test queries at ~2–4 second average response time, automating natural language data extraction from a 1,000+ row structured dataset

- Implemented a LangGraph agent with 2 automated tool-call routing functions, replacing manual query handling with a fully automated, governance-aware data retrieval workflow
- Containerised the full application using Docker and docker-compose, enabling reproducible deployment and consistent environment management across development and production
- Incorporated privacy-aware data handling practices throughout API design, ensuring sensitive data fields were correctly identified, categorised, and handled in line with data protection principles
- Maintained full codebase under Git/GitHub with CI/CD integration, supporting automated validation checks and continuous compliance across all pipeline components

Energy Finance Data Engineering & Analytics Platform

- Consolidated **20,000+ energy and financial records** from **4 source datasets**, measured by successful automated ingestion and cleansing, by building a **cloud ETL pipeline** with **Azure Data Factory, ADLS, and Databricks (Python/SQL)**
- Collaborated in an **Agile/Scrum team structure**, measured by consistent sprint delivery across the project timeline, by participating in **sprint planning, standups, and iterative development cycles**
- Structured a **5-table star schema (2 fact, 3 dimension tables)**, measured by query-ready analytical structure, by applying **dimensional data modelling** and loading curated data into **Azure Synapse SQL Pool**
- Delivered **10+ reusable financial and energy KPIs**, measured by production, revenue, cost, budget, and variance analysis coverage, by writing **SQL queries** against the **Synapse and Snowflake warehouse layer**
- Enabled **decision-ready reporting**, measured by **10+ KPIs** visualized for actual-vs-budget and plant performance tracking, by connecting the warehouse to interactive **Power BI dashboards**

Automated Music Trend Pipeline (Airflow + dbt + PostgreSQL)

- Consolidated 10,000+ Spotify records ingested daily from multiple sources into a migration-ready PostgreSQL store by designing an end-to-end data pipeline with Apache Airflow.
- Reduced query complexity by ~60% by developing 3 structured dbt transformation models—genre_summary, artist_trends, and decade_analysis—to produce consistent, query-ready datasets.
- Established a clean, well-documented data foundation across 3 analytical domains by applying structured data modeling and hygiene standards throughout the ingestion, transformation, and loading process.
- Automated stakeholder-ready reporting by connecting pipeline outputs to a Looker Studio dashboard, enabling real-time refreshes with zero manual export effort.